

Cryptography I: Lecture Notes

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1 Applying a function to a random variable

When describing a probabilistic computation, we often apply a deterministic function $f : \mathcal{X} \rightarrow \mathcal{Y}$ to a random input X . Operationally, in each run we take the value x of X and compute $f(x)$. Since the result depends on X , it is itself a random variable, which we denote by $f(X)$. To connect this notation with the formal definition of random variables, recall that a random variable is a function $X : \Omega \rightarrow \mathcal{X}$, where Ω is the sample space. Each outcome $\omega \in \Omega$ determines an input value $X(\omega)$, and hence an output value $f(X(\omega))$. Thus, $f(X)$ is the *composition*

$$f(X) := f \circ X : \Omega \rightarrow \mathcal{Y}, \quad \omega \mapsto f(X(\omega)).$$

In other words, $f(X)$ is shorthand for applying f to the value of X at each outcome. Note that X and $f(X)$ share the same underlying randomness: computing $f(X)$ involves no new random choices.¹

This notation appears, for example, in Exercise 1.5 (*How function application changes statistical distance*), which asks us to show that

$$\Delta(f(A) ; f(B)) \leq \Delta(A ; B).$$

Here, $f(A)$ and $f(B)$ are obtained by applying the same function f to the values of A and B , respectively, so the inequality compares how distinguishable the two distributions are before and after this transformation. Intuitively, applying f can only lose information: f may merge distinct inputs into a single output, which can never make the distributions easier to tell apart.

¹ On general measurable spaces, $f(X)$ is a random variable whenever f is measurable.